Scientific Justification *Be sure to include overall significance to astronomy. For standard proposals limit text to one page with figures, captions and references on no more than two additional pages.*

Quasi-Periodic Eruptions (QPEs) are intense, recurring, rapid bursts of soft X-ray radiation originating from the nuclei of galaxies, typically recurring every few hours. These events are thought to be prime candidates for the electromagnetic counterparts to Extreme Mass-Ratio Inspirals (EMRIs), stellar-mass compact objects orbiting supermassive black holes (SMBHs), which are a key source class for future low-frequency gravitational wave (GW) detectors like LISA. However, the physical origin of QPEs is fiercely debated. Competing models, such as repeating disk instabilities or accretion impacts from an orbiting body, make nearly identical predictions for their X-ray spectra and coarse timing properties.

Since the X-ray spectra are not enough, we must use the optical light curve to delineate our models. If QPEs are caused by an EMRI directly punching through the accretion disk, we will recover strictly periodic eruptions, with the quasi-periodic timing deviations (for example: from gravitational radiation or apsidal precession) occurring on predictable, long timescales. In contrast, a disk instability model, where a star orbiting a SMBH undergoes mass transfer, might showcase quasi-periodicity by default due to the nature of the mass transfer's fluid flow, with stochastic jitter or systematic period evolution on much shorter timescales. Current optical observations of QPEs are sparse and lack the continuous, high-cadence coverage needed to perform these precise timing analyses. The absence of this data is the primary obstacle to determining the nature of QPEs and, by extension, their utility as multi-messenger beacons for EMRIs.

We propose a new experiment to break this model degeneracy: obtaining the first continuous, high-cadence optical light curve of a QPE throughout multiple eruption cycles. By measuring the arrival times of optical eruptions with minute-level precision for multiple connected cycles, we can detect or constrain period derivatives ($\dot{P}$) and timing jitter. A null detection of significant stochastic jitter or a $\dot{P}$ consistent with GW-driven inspiral would strongly favor the EMRI hypothesis. On the other hand, detection of significant jitter or an anomalous $\dot{P}$ would point toward accretion disk physics as the driver. This experiment directly addresses a fundamental question in high-energy astrophysics and while opening the door for a new EM-GW observation synergy within LISA's science case.

References

Arcodia et al. 2021, *Nature*, 592, 704 (QPE Discovery in eROSITA data)

Miniutti et al. 2023, *A&A*, 670, A93 (Detailed study of GSN 069)

Linial & Metzger 2023, *ApJ*, 957, 34 (EMRI model for QPEs)

Krolik & Linial 2022, *ApJ*, 941, 24 (Disk instability model for QPEs)

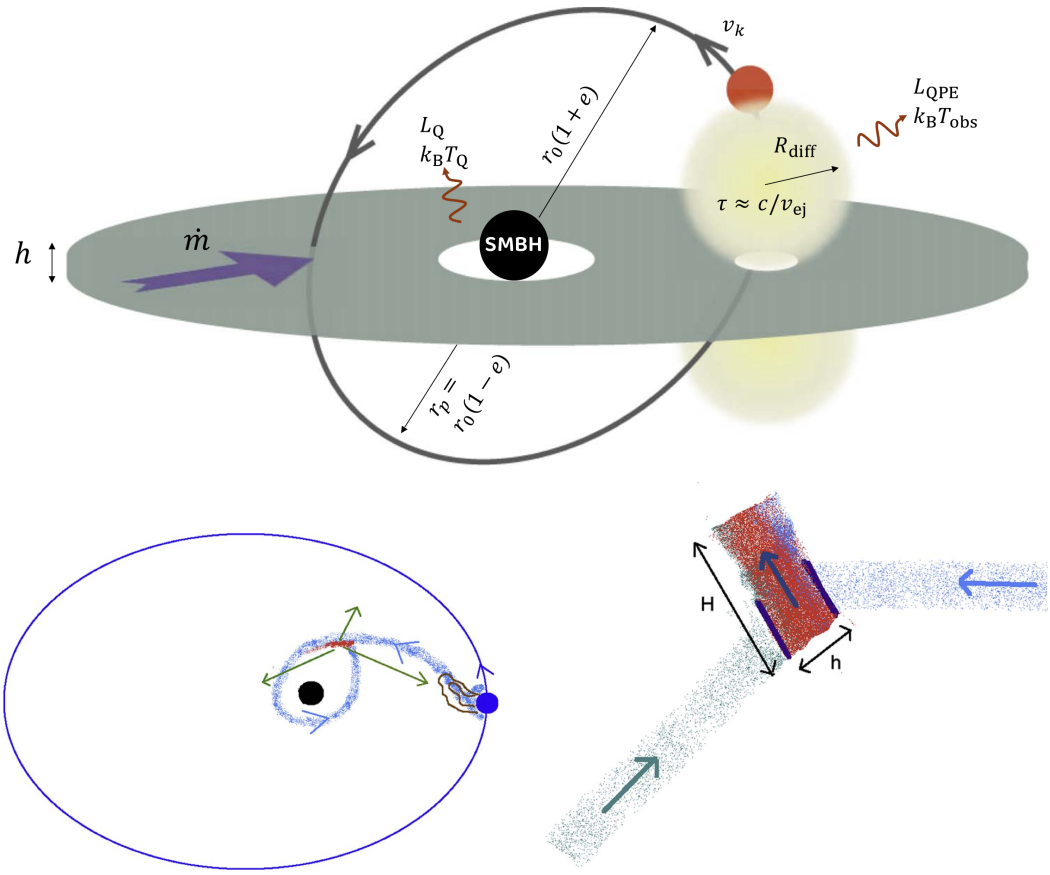

Figure 1: Examples from Linial&Metzger+23 and Krolik&Linial+22 (see both papers for more in-depth description) of an EMRI vs a disk instability model. In the EMRI model (above), the EMRI is repeatedly punching through the the disk, with the orbital shrinking from friction with the disk at the points of crossing causing the orbit's shrinking. Meanwhile, the disk instability case has a star losing its mass to the black hole, and causing shocks further in the gravitational well which cause the eruptions (by the mass transfer's non-linear nature, these shocks are quasi-periodic). GSN 069 is perfect to test the differences between the models, since the secondary object has a known optical counterpart, which means it is tidally-deformable and therefore likely to leech mass. The question this observing proposal wishes to answer is to determine which of these models dominate the EMRI signature. If the disk instability model is preferred, then tidal deformability is a must for QPEs to happen, meaning that for black holes, which the most likely EMRI secondary object, we will not have an electromagnetic signal to work with to help with parameter estimation.

Experimental Design *Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification?*

We will execute a targeted and intensive monitoring campaign designed to achieve the precise timing measurements required for differentiating the two mentioned models. GSN 069 will be our target since it has a well-characterized recurrence period of $P_{\rm rec} \approx 9$ hours, a burst timing of $T_{\rm burst} \lesssim 30$ minutes. Its optical counterpart is a known Seyfert 2 galaxy (2dFGRS TGS564Z081) with magnitudes of $B = 16.2$ and $R = 15.3$, which is more than adequate for our high-cadence photometry plan with 1m telescopes. We will observe in the SDSS-r(p) band, since it is the preferred band for observing AGN activity in the literature due to optimal balancing of sensitivity and resolution.

The core of our strategy is to obtain high-cadence photometry around the GSN 069's predicted bursts, sampled with a cadence $T_c \approx 5min$ for spans of 2 hours, which will allow us to refine the imaging of GSN 069 more than any other survey since. We request two primary observing epochs for our two-hour windows, each spanning around 100 hours (covering $9 - 10$ eruption cycles), separated by approximately one month: the first establishing the short-term timing precision, while the second searches for the long-term period derivative predicted by EMRI models.

The novel capability we leverage is the global telescope network of the Las Cumbres Observatory (LCO). Since it is moves across the southern sky throughout the day, we will observe at Cerro Tololo (Chile), the South African Astronomical Observatory site, and also the Siding Spring (Australia) site to maximize our cadence and observations. We will provide precise timings and sky locations for these observations by the end of this proposal drafting process.

Using the Las Cumbres Observatory's (LCO) Exposure Time Calculator with a conservative target magnitude ($r = 16$), we determine that three minute exposures (3 min exposure + 2 min processing time) on a 1m telescope yield a SNR ≈ 135 per exposure. According to our SNR calculations (see figure 2), this will be just enough to distinguish jitter timescales down to three minutes, which is more than any study in the literature has been capable of.

References

Colless et al. 2001, *MNRAS*, 328, 1039-1063 (The 2dF Galaxy Redshift Survey: spectra and redshifts)

Shalyapin et al. 2021, *A&A*, 646, A165 (Resolving the inner accretion flow towards the central SMBH in SDSS J1339+1310)

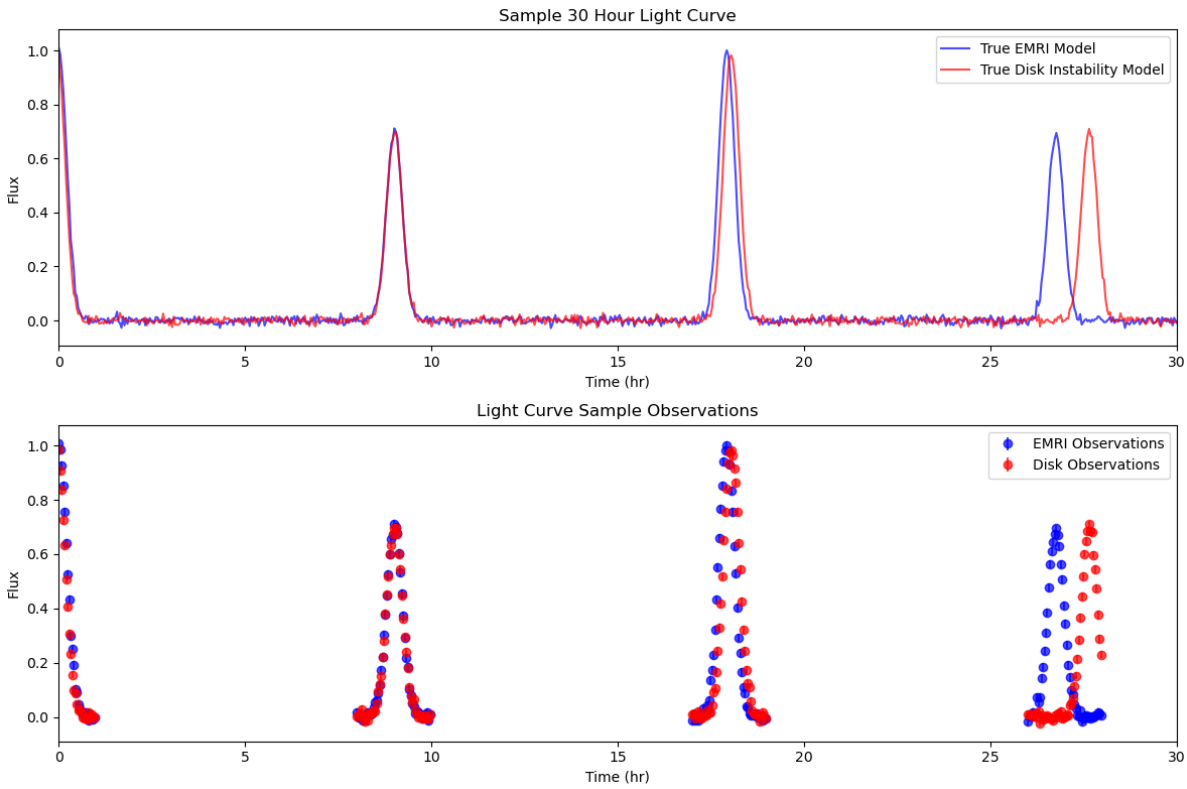

Figure 2: Example light curve and sampling plan for distinguishing the EMRI and disk instability models. The SNR required to distinguish these models is related to the jitter w.r.t. the EMRI model that the disk instability model experiences. This plot was generated with a 9 hour period, 30 minute burst time (Gaussian burst), 5 minute/period derivative, 2 hour observation windows with $\approx 30 - 50$ samples, and a 24 minute jitter in the disk instability model. The jitter will likely be much smaller than that, but we can approximate the SNR derived from the jitter with SNR $\approx \frac{\Delta t_{\rm jitter}\sqrt{N_{\rm peaks}}}{T_{\rm c}}$, where $\Delta t_{\rm jitter}$ is the jittering scale, $N_{\rm peaks}$ is the number of observed peaks, and $T_{\rm c}$ is how often we take observations (cadence). At $r = 16$ mag, the max SNR within a five minute exposure tops out below 200, which is insufficient to resolve minute-scale jitters, but enough to resolve below five.